CHAPTER ONE

1.0 MEASUREMENT

Measurement is very important to all fields of study and it cuts across all areas of live. The technological advancement of any nation depends largely on how accurately it can measure physical and electrical quantities. Measurement can simply be referred to as a process of finding the amount or magnitude of a physical quantity. It can also be expressed as a means of detecting or obtaining the amount of unknown physical quantity using known or standard physical quantity. The systems of measurement include S.I. unit, metric unit, British system and United State customary system. Measurement can be made directly or indirectly.

Direct Measurement: This can be done by bringing the target physical quantity into contact with the measurement system.

Indirect Measurement: This can be achieved by comparing the magnitude of the quantity being measured with a reference or known value.

1.1 Measuring instrument

A **measuring instrument** is a device used to measure a physical quantity. In the physical sciences, quality assurance, and engineering, measurement is the activity of obtaining and comparing physical quantities of real-world objects and events. Established standard objects and events are used as units, and the process of measurement gives a number relating the item under study and the referenced unit of measurement. Measuring instruments, and formal test methods which define the instrument's use, are the means by which these relations of numbers are obtained. All measuring instruments are subject to varying degrees of instrument error and measurement uncertainty. These instruments may range from simple objects such as rulers and stopwatches to electron microscopes and particle accelerators. Virtual instrumentation is widely used in the development of modern measuring instruments.

1.2 **Types of Measuring Instruments and Their Functions**

A measuring instrument is a tool used to measure the unit value of a quantity. The quantities here vary, such as weight, temperature, length, time, and so on. For example, to measure the length of a branch, you can measure it by using a ruler, which means the quantity measured here is the length and the measuring instrument is a ruler. Measuring instruments that are wrong in their use will result in measurement errors. For this reason, new users of certain measuring instruments need to read the guidelines for using measuring instruments first. There are various types of measuring instruments that have their respective functions. They can be classified into analog and digital instruments. Some measuring instruments are commonly used by the general public and some other measuring instruments are used by a certain group of people only. In this article, various types of measuring instruments and their functions will be presented along with pictures so that you will know more about measuring instruments that you may use someday.

1.2.1 Ruler

A ruler is a long measuring instrument that is used to measure the length of an object. Generally, rulers are made of iron, mica or wood. This measuring instrument is commonly used by the community and is even considered a necessity for children at school. There are 10 cm- to 100 cm-length rulers. The most common length for a ruler is 30 cm.

1.2.2. Roll Meter

The roll meter is an instrument for measuring the length of an object. This tool is similar to a ruler, except that the roll meter has a longer measuring distance of up to 50 meters. This tool is used in the furniture industry to measure wood or planks for furniture. There is also a roll meter measuring up to 100 meters.

1.2.3. Vernier Calipers

A caliper is an instrument used to measure the length, thickness, diameter, and depth of an object. It has a measurement accuracy of 0.1 mm.

1.2.4. Micrometer Screw Gauge

A micrometer is also used to measure the diameter and thickness of an object. However, this tool has a higher accuracy than calipers with an accuracy of 0.01 mm. Therefore a micrometer is used to measure the diameter and thickness of smaller objects, such as coins.

1.2.5. Scales

The scale is a measuring instrument used to measure the weight or mass of an object. There are various kinds of scales such as sitting scales, body scales, three-arm scales, digital scales, hanging scales, and others. The scale in the image is a type of sitting scale that is generally used to measure the weight of food ingredients and other commercial objects. Scales are also available in various units depending on the type.

1.2.6. Thermometer

A thermometer is an instrument that can measure temperature. The temperature read in a thermometer is degree Celsius (^{0}C). There are several types of thermometers. The picture below is an example of the most commonly used type of thermometer in hospitals or in the laboratory. There is also a type of thermometer called infrared thermometer that is used to measure body temperature from a relatively short distance.

1.2.7. Beaker Glass,

Beaker glass is a volume measuring instrument for measuring the volume of a liquid. This tool is generally used in the laboratory and is used to measure the volume of liquid chemicals before being researched. On the beaker glass, there is a scale in the form of a line that shows the volume size. This tool is available in various sizes, usually the unit is ml.

1.2.8. Stopwatch

A stopwatch is used to measure time in seconds. This instrument is usually used to determine runner speed, reaction time, heating time, or time to work on something that is under one hour. This instrument is sometimes replaced by a watch, but the accuracy of the stopwatch is higher than that of a watch.

1.2.9. Barometer

Barometer is used to measure the pressure of a place or room. This tool is placed in a chemistry or biology laboratory which helps determine the pressure when there is a reaction or the planting of bacteria that is affected by pressure. The instrument can also be utilized in altitude measurement.

1.2.10. Voltmeter

As the name suggests, a voltmeter is used to measure the amount of electric voltage in units of volts. The method of measurement is by connecting the voltmeter in parallel with the device, then the pointer will deflect according to the electrical voltage of the device.

1.2.11. Ammeter

The ammeter is almost the same as a voltmeter. Its shape is almost the same, too. However, this instrument is used to measure the electric current of a circuit in amperes and must be connected in series with in the circuit.

1.2.12. Dial Indicator

A dial indicator is used to measure the flatness of a flat area. It is generally used in the manufacturing industry to measure the flatness of tools, as well as in workshops which are usually used to measure the flatness of the disc brake in a car. This instrument has an accuracy of 0.01 mm.

1.2.13. Sound Level Meter

A sound level meter is used to measure the noise level of a place. It is generally used in noise-prone areas such as in industry, aviation, shooting training locations, and so on. Noise needs to be measured to ensure that in such places it is within normal limits because noise damages the eardrum.

1.2.14. Manometer

A manometer is used to measure gas pressure by connecting it to the gas containing vessel using a pipe or tube. This tool is used in industries that run on gas or industries that produce gas.

1.2.15. Densitometer

This instrument is used to measure the amount of light absorbed by an object or the level of darkness of a semitransparent object or in the form of a film. It can be used simply by pressing the reset button and placing the object on the densitometer. Then, the darkness value will appear on the display.

1.2.16. Hygrometer

Hygrometer is a measuring instrument used to measure the humidity of a room. Usually, this instrument is used in places that pay attention to humidity such as in the food industry, to prevent high humidity which results in the growth of mold on food.

1.2.17. pH Meter

pH meter is used to measure the pH of a material. pH here is the level of acidity or alkalinity of a material. It can be used simply by dipping the pH meter electrode (which is shaped like a pen) into the material.

1.2.18. Anemometer

Anemometer is a measuring instrument for the wind speed of a place. The speed measurement is then used by the geophysical meteorological agency or weather forecasting agency to predict the future weather. The bowl-shaped propeller will rotate when it gets windy and then it is converted by the instrument into a speed value.

1.3 Measurement Error

Error may arise in a measurement as result of many factors. The measurement error is defined as the difference between the true or actual value and the measured value. It can also be referred to as the measure of deviation of measured value from the standard value.

Types of Errors in Measurement

The error may arise from the different source and are usually classified into the following types. These types are

1. Gross Errors
2. Systematic Errors
3. Random Errors

1.3.1. Gross Errors: The gross error is a class of error which may occur as a result of the human mistakes. For examples consider the person using the instrument takes the wrong reading or record the incorrect data, such type of error comes under the gross error. The gross error can only be avoided by taking the readings carefully and repeatedly. Two or more readings should be taken of the measurement quantity. The readings are taken by the different observers and at a different points to remove the error.

1.3.2. Systematic Errors: The systematic errors are mainly classified into three categories. This error is closely associated/instrument being used for measurement. It consistently affects measurements in the same direction. It is sub-divided into;

a. Instrumental Errors
b. Environmental Errors
c. Observational Errors

(a) Instrumental Errors: These errors mainly arise due to the three main reasons.

(i) Inherent Shortcomings of Instruments – Such types of errors are inbuilt in instruments because of their mechanical structure. They may be due to manufacturing, calibration or operation of the device. These errors may cause the measuring instrument to read too low or too high.

For example – If the instrument uses the weak spring then it gives the high value of measuring quantity. The error occurs in the instrument because of the friction or hysteresis loss.

(ii) Misuse of Instrument – The error occurs in the instrument because of the fault of the operator. A good instrument used in an unintelligent way may give an enormous result.

For example – the misuse of the instrument may cause the failure to adjust the zero of instruments, poor initial adjustment, using lead to too high resistance. These improper practices may not cause permanent damage to the instrument, but all the same, they cause errors.

(iii) Loading Effect – It is the most common type of error which is caused by the instrument in measurement work. For example, when the voltmeter is connected to the high resistance circuit it gives a misleading reading, and when it is connected to the low resistance circuit, it gives the dependable reading. This means the voltmeter has a loading effect on the circuit.

The error caused by the loading effect can be overcome by using the meters intelligently. For example, when measuring a low resistance by the ammeter-voltmeter method, a voltmeter having a very high value of resistance should be used.

(b) Environmental Errors

These errors are due to the external condition of the measuring devices. Such types of errors mainly occur due to the effect of temperature, pressure, humidity, dust, vibration or because of the magnetic or electrostatic field. The corrective measures employed to eliminate or to reduce these undesirable effects are

- The arrangement should be made to keep the conditions as constant as possible.
- Using the equipment which is free from these effects.
- By using the techniques which eliminate the effect of these disturbances.
- By applying the computed corrections.

(c) Observational Errors

Such types of errors are due to the wrong observation of the reading. There are many sources of observational error. For example, the pointer of a voltmeter resets slightly above the surface of the scale. Thus an error **occurs** (because of parallax) unless the line of vision of the observer is exactly above the pointer. To minimize the parallax error highly accurate meters are provided with mirrored scales.

1.3.3. Random Errors

This type of error is caused by the sudden and unpredictable change in either direction. These types of error remains even after the removal of the systematic error. Hence such type of error is also called residual error. The error occurs by chance. The error can be minimized by understanding the theoretical background of the experiment and studying instrument manual carefully.

1.4 Terms Associated with Measurement

The commonly used terms are accuracy, precision, calibration, resolution, e.t.c.

1. Accuracy – The degree of exactness (closeness) of a measurement compared to the expected (desired) value. The quantity being measured is said to be accurately measured if it has the same value as the standard value.
2. Precision – This is a measure of consistency or repeatability of measurements, i.e. successive readings do not differ or the consistency of the instrument output for a given value of input. A very precise measuring instrument may not perfectly give an accurate reading.
3. Calibration – This refers to the series of operations aimed at ascertaining the relationship/difference between actual equipment value and reference value.

1.5 How to Reduce Errors in Measuring Instruments

Error in measuring instrument can be reduced by;

1. Calibration
2. Carrying out regular maintenance
3. Narrowing down tolerance range
4. Using standard measuring techniques
5. Training
6. Adopting automotive measurement system
7. Studying instrument manual
8. Repairing faulty parts
9. Proper installation
10. Storing it in good environment.

1.6 Care of Electronics Measuring Instruments

- Store instruments in the appropriate location
- Lubricate instruments properly to avoid corrosion
- Ensure proper handling of instruments
- Avoid exposing them to temperature extremes
- Obtain professional maintenance services if needed
- Clean after every use
- Use them for their designated purposes only

CHAPTER TWO

2.0 ELECTRONIC MEASURING INSTRUMENT

These are instruments which employ electronic devices and circuitry for measuring physical quantities. Examples of electrical quantities that can be measured with the instruments are voltage, current, resistance, charge, e.t.c. Although certain measurement can be carried out by mechanical or electrical means, measurement done electronically tend to have higher sensitivity, better stability, greater flexibility and easy readability. They can be grouped into analog and digital instrument. A typical measuring instrument consists of various units as discussed below:

1. Transducer: A transducer is a device that changes the physical quantity to be measured into its equivalent electrical form. The applied input can be temperature, pressure, velocity, displacement etc. It basically converts a form of energy into another. Meanwhile not all measuring instrument require transducers especially whe the quantity to be measured is already in electrical form. Examples are LDR, thermistor, e.t.c.
2. Signal processing unit: This unit is mainly composed of amplification circuitry along with balancing circuits and calibrating elements. The output of the transducer which is the electrical form of the quantity to be measured is fed to the signal processing unit/signal modifier.This basically amplifies or modifies the output of the transducer so that it can be easily detected and accepted by the other units of the system. The signal modifier could be analog to digital converter, signal converter, multiplexer, amplifier, etc.

Multiplexer: Multiplexer mixes the multiple analog signals supplied by the processing unit. It produces an individual signal by muxing various applied signals. This signal is then processed further.

Signal converter: This unit takes the output of the multiplexer and generates such a signal that can be processed by further units of the system.

Analog to digital converter: ADC plays a crucial role in the digital instrumentation system. It converts the applied analog data signal into its equivalent digital form. This converted digital data is then provided to the display unit.

3. Display unit: It is the part of electronic measuring instrument where the magnitude of the quantity being measured is displayed. This unit shows the actual quantity that is to be measured in the numerical form. This can be a CRO or a computer monitor etc. depending on the need of the user.

2.1 Analog Instrument

The analogue instrument is defined as the instrument whose output is the continuous function of time, and they have a constant relation to the input. The physicals quantity like voltage, current, power and energy are measured using analogue instruments. Most of the analogue instrument use pointer or dial which move over a scale to indicate the magnitude of the measured quantity. The degree of deflection of the pointer is a function of the magnitude of the measurand. Examples are voltmeter, ammeter, galvanometer, e.t.c.

2.2 Digital Instruments

The instruments that are used to express the measuring quantity in numeric/alphabetic format are known as Digital Instruments. They use logic circuits and techniques to obtain measurement and then display it in form of numbers or alphabets. Digitized information is somewhat easy to be handled and transmitted thus widely preferred nowadays. The digital readout employs either LED or LCD.

Quantization is the basis of working of a digital instrument. It is an act of transforming an analog signal into its digital form. Digital instruments are composed of logic circuits that carry out measurement of the quantities.

As a result of several advantages of digital instruments such as high speed, errorless results, better resolution and greater accuracy over analog instruments, the popularity of digital instruments are increasing rapidly.

Basic Digital instruments

- **Digital Voltmeter**:

A digital voltmeter is a device that measures voltage and displays the quantity directly in a digital manner rather than analog representation. Here, the measured voltage is provided in the form of discrete numerals. It is abbreviated as DVM.

Due to its versatility and accuracy, it is widely used in various ac and dc voltage measuring applications.

- **Digital multimeter**:

A digital multimeter helps us to measure voltage, current and resistance in a digital manner. Or we can say, that a digital multimeter is a type of digital instrument that generates a digitized value of voltage, current or resistance. DMM is the acronym for digital multimeter.

As digital instruments are said to be highly accurate but it is noteworthy in their case that the level of accuracy somewhat depends on the analog to digital converter employed in the system. A key factor that differentiates the digital voltmeter from the analog one, is the high input impedance offered by it along with small instrument size.

- **Digital frequency meter**:

It is a type of digital instrument that measures the frequency of the applied periodic signal. It consists of an attenuator, wave-shaping circuit along with logic gates, counters and flip-flops in order to measure the frequency of the electrical signal.

Digital frequency meter measures the signal frequency by converting it into a train of pulses. In other words, a combination of pulses is the applied input to the system. Here, each cycle of the signal to be measured represents one pulse.

The pulses that occur on definite time duration are then counted. This count shows the frequency of the applied input signal. The pulse count is done through an electronic counter having high speed. Due to this, signals having high frequency can also be measured.

2.3 Permanent Magnet Moving Coil (PMMC) Instrument

The instruments which use the permanent magnet for creating the stationary magnetic field between which the coil moves is known as the permanent magnet moving coil or PMMC instrument. It operates on the principle that the torque is exerted on the moving coil placed in the field of the permanent magnet. The PMMC instrument gives the accurate result for DC measurement.

2.3.1 Construction of PMMC Instrument

The moving coil and permanent magnet are the main part of the PMMC instrument.
The parts of the PMMC instruments are explained below in details.

Moving Coil – The coil is the current carrying part of the instruments which is freely moved between the stationary field of the permanent magnet. The current passes through the coil deflects it due to which the magnitude of the current or voltage is determined. The coil is mounted on the rectangular former which is made up of aluminium. The former increases the radial and uniform magnetic field between the air gap of the poles. The coil is wound with the silk cover copper wire between the poles of a magnet.

Magnet System – The PMMC instrument uses the permanent magnet for creating the permanent magnetic field. The Alcomax and Alnico material are used for creating the permanent magnet because this magnet has the high coercive force. Also, the magnet has high field intensities.

Control – In PMMC instrument the controlling torque is provided by the springs. The springs are made up of phosphorous bronze and placed between the two jewel bearings. The spring also provides the path through which current flow in and out of the moving coil. The controlling torque is mainly because o\f the suspension of the ribbon.

Damping – The damping torque is used for keeping the movement of the coil in rest. This damping torque is induced because of the movement of the aluminium core which is moving between the poles of the permanent magnet.

Pointer & Scale – The pointer is linked with the moving coil. The pointer notices the deflection of the coil, and the magnitude of its deflection is shown on the scale. The pointer is made of the lightweight material, and hence it is easily deflected with the movement of the coil. Sometimes the parallax error occurs in the instrument which is easily reduced by correctly aligning the blade of the pointer.

Torque Equation for PMMC Instrument

The deflecting torque(T_d) induces because of the movement of the coil. The deflecting torque is expressed by the equations 2.1 and 2.2 below.

$$Td = NBAI \qquad (2.1)$$

$$\therefore Td = GI \qquad (2.2)$$

Where,

N – Number of turns of coil
B – flux density in the air gap(Wb/m^2)
A – effective area of the coil (m^2)
I – current through the coil (A)

The spring provides the restoring torque to the moving coil which is expressed as

$$Tr = k\theta \qquad (2.3)$$

Where, $\theta = angular\ deflection$ and k = Spring constant(Nm/Deg).

For final deflection, $Td = Tr$

And $$\theta = \left(\frac{G}{k}\right) I \qquad (2.4)$$

The above equation shows that the deflection torque is directly proportional to the current passing through the coil.

2.3.2 Error in PMMC Instruments

In PMMC instruments the error occurs because of the ageing and the temperature effects of the instruments. The magnet, spring and the moving coil are the main parts of the instruments which cause the error. The different types of errors of the instrument are explained below in details.

1. Magnet – The heat and vibration reduce the lifespan of the permanent magnet. This treatment also reduced the magnetism of the magnet. The magnetism is the property of the attraction or repulsion of the magnet. The weakness of the magnet decreases the deflection of the coil.

2. Springs – The weakness of the spring increases the deflection of moving coil between the permanent magnet. So, even for the small value of current, the coil show large deflection. The spring gets weakened because of the effect of the temperature. One degree rise in temperature reduces the 0.004 percent life of the spring.

3. Moving Coil – The error exists in the coil when their range is extended from the given limit by the use of the shunt. The error occurs because of the change of the coil resistance on the shunt resistance. This happens because the coil is made up of copper wire which has high shunt resistance and the shunt wire made up of Magnin has low resistance.

To overcome from this error, the swamping resistance is placed in series with the moving coil. The resistor which has low-temperature coefficient is known as the swamping resistance. The swamping resistance reduces the effect of temperature on the moving coil.

2.3.4 Advantages of PMMC Instruments

The following are the advantages of the PMMC Instruments.

1. The scale of the PMMC instruments is correctly divided.
2. The power consumption of the devices is very less.
3. The PMMC instruments have high accuracy because of the high torque weight ratio.
4. The single device measures the different range of voltage and current. This can be done by the use of multipliers and shunts.
5. The PMMC instruments use shelf shielding magnet which is useful for the aerospace applications.
6. It has no hysteresis loss
7. It has efficient eddy current damping
8. It possesses high torque-to-weight ratio

2.3.5 Disadvantages of PMMC Instruments

The following are the disadvantages of the PMMC instruments.

1. The PMMC instruments are only used for the direct current. The alternating current varies with the time. The rapid variation of the current varies the torque of the coil. But the pointer cannot follow the fast reversal and the deflection of the torque. Thus, it cannot be used for AC.
2. The cost of the PPMC instruments is much higher as compared to the moving coil instruments.

The moving coil itself provides the electromagnetic damping. The electromagnetic damping opposes the motion of the coil which is because of the reaction of the eddy current and the magnetic field.

2.4 Moving Coil Galvanometer

A galvanometer is a device that is used to detect small electric current or measure its magnitude. The current and its intensity is usually indicated by a magnetic needle’s movement or that of a coil in a magnetic field that is an important part of a galvanometer.

Some of the different types of galvanometer include Tangent galvanometer, Astatic galvanometer, Mirror galvanometer and Ballistic galvanometer. However, today the main type of galvanometer type that is used widely is the D’Arsonval/Weston type or the moving coil type. A galvanometer is basically a historical name that has been given to a moving coil electric current detector.

A moving coil galvanometer is an instrument which is used to measure electric currents. It is a sensitive electromagnetic device which can measure low currents even of the order of a few microamperes.

Moving-coil galvanometers are mainly divided into two types:

- Suspended coil galvanometer
- Pivoted-coil or Weston galvanometer

2.4.1 Moving Coil Galvanometer Principle

A current-carrying coil when placed in an external magnetic field experiences magnetic torque. The angle through which the coil is deflected due to the effect of the magnetic torque is proportional to\ the magnitude of current in the coil.

2.4.2 Moving Coil Galvanometer Construction and Diagram

The moving coil galvanometer is made up of a rectangular coil that has many turns and it is usually made of thinly insulated or fine copper wire that is wounded on a metallic frame. The coil is free to rotate about a fixed axis. A phosphor-bronze strip that is connected to a movable torsion head is used to suspend the coil in a uniform radial magnetic field. Essential properties of the material used for suspension of the coil are conductivity and a low value of the torsional constant. A cylindrical soft iron core is symmetrically positioned inside the coil to improve the strength of the magnetic field and to make the field radial. The lower part of the coil is attached to a phosphor-bronze spring having a small number of turns. The other end of the spring is connected to binding screws. The spring is used to produce a counter torque which balances the magnetic torque and hence helps in producing a steady angular deflection. A plane mirror which is attached to the suspension wire, along with a lamp and scale arrangement, is used to measure the deflection of the coil. Zero-point of the scale is at the centre.

2.4.3 Working of Moving Coil Galvanometer

Let a current I flow through the rectangular coil of n number of turns and a cross-sectional area A. When this coil is placed in a uniform radial magnetic field B, the coil experiences a torque τ.

Let us first consider a single turn ABCD of the rectangular coil having a length l and breadth ***b*.** This is suspended in a magnetic field of strength B such that the plane of the coil is parallel to the magnetic field. Since the sides AB and DC are parallel to the direction of the magnetic field, they do not experience any effective force due to the magnetic field. The sides AD and BC being perpendicular to the direction of field experience an effective force F given by

$$F = BIl \qquad (2.5)$$

Using Fleming's left-hand rule we can determine that the forces on AD and BC are in opposite direction to each other. When equal and opposite forces F called couple acts on the coil, it produces a torque. This torque causes the coil to deflect.

We know that torque τ = force x perpendicular distance between the forces

$$\tau = F \times b \qquad (2.6)$$

Substituting the value of F we already know,

Torque τ acting on single-loop ABCD of the coil = $BIl \times b$

Where lx b is the area A of the coil,

Hence the torque acting on n turns of the coil is given by

$$\tau = nIAB \qquad (2.7)$$

2.4.4 Sensitivity of Moving Coil Galvanometer

The general definition of the sensitivity experienced by a moving coil galvanometer is given as the ratio of change in deflection of the galvanometer to the change in current in the coil.

$$S = d\theta/dI \qquad (2.8)$$

The sensitivity of a galvanometer is higher if the instrument shows larger deflection for a small value of current. Sensitivity is of two types, namely current sensitivity and voltage sensitivity.

- **Current Sensitivity**

The deflection θ per unit current I is known as current sensitivity θ/I

$$\theta/I = nAB/k \qquad (2.9)$$

- **Voltage Sensitivity**

The deflection θ per unit voltage is known as Voltage sensitivity θ/V. Dividing both sides by V in the equation $\theta = (nAB / k)I$;

$\theta/V = (nAB / V k)I = (nAB / k)(I/V) = (nAB / k)(1/R)$

R stands for the effective resistance in the circuit.

It is worth noting that voltage sensitivity = Current sensitivity/ Resistance of the coil. Therefore, under the condition that R remains constant; voltage sensitivity ∝ Current sensitivity.

2.4.5 Factors Affecting Sensitivity Of A Galvanometer

a) Number of turns of the coil

b) Area of the coil

c) Magnetic field strength B

d) The magnitude of couple per unit twist k/nAB

2.4.6 Applications of Galvanometer

The moving coil galvanometer is a highly sensitive instrument due to which it can be used to detect the presence of current in any given circuit. If a galvanometer is a connected in a Wheatstone’s bridge circuit, the pointer in the galvanometer shows null deflection, i.e no current flows through the device. The pointer deflects to the left or right depending on the direction of the current.

The galvanometer can be used to measure:

a) the value of current in the circuit by connecting it in parallel to low resistance.

b) the voltage by connecting it in series with high resistance.

2.5 Conversion of Galvanometer to Ammeter

A galvanometer is converted into an ammeter by connecting it in parallel with a low resistance called shunt resistance. Suitable shunt resistance is chosen depending on the range of the ammeter.

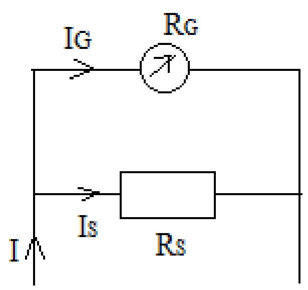


Fig 2.1

In the given circuit of figure 2.1,

R_G – Resistance of the galvanometer

G- Galvanometer coil

I – Total current passing through the circuit

I_G – Total current passing through the galvanometer which corresponds to full-scale reading

R_s – Value of shunt resistance

When current I_G passes through the galvanometer, the current through the shunt resistance is given by $I_S = I - I_G$. The voltages across the galvanometer and shunt resistance are equal due to the parallel nature of their connection.

Therefore $R_G \cdot I_G = (I - I_G) \cdot R_s$

The value of R_s can be obtained using the above equation.

2.6 Conversion of Galvanometer to Voltmeter

A galvanometer is converted into a voltmeter by connecting it in series with high resistance. A suitable high resistance is chosen depending on the range of the voltmeter.

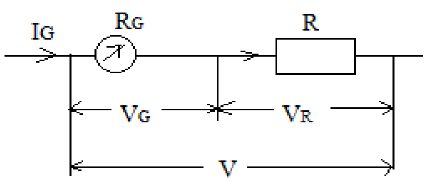


Fig 2.2

In the given circuit 2.2,

R_G = Resistance of the galvanometer

R = Value of high resistance

G = Galvanometer coil

I = Total current passing through the circuit

I_G = Total current passing through the galvanometer which corresponds to a full-scale deflection.

V = Voltage drop across the series connection of galvanometer and high resistance.

When current I_G passes through the series combination of the galvanometer and the high resistance R; the voltage drop across the branch is given by

$V = R_G.I_G + R.I_G$

The value of R can be obtained using the above equation.

2.7 Worked Examples

Question1: A moving coil galvanometer of resistance 100Ω is used as an ammeter using a resistance of 0.1Ω. The maximum deflection current in the galvanometer is 100µA. Find the current in the circuit, so that the ammeter shows maximum deflection.

Solution: It is given that R_G =100Ω , R_s= 0.1Ω, I_G =100µA

We know that $R_G .I_G = (I - I_G).R_S$

Therefore $I = (R_G .I_G + I_G.R_s)/ R_S$

$I = (1+R_G/ R_S). I_G$

Substituting the given values, we get I = 100.1mA

Question2: A galvanometer coil of 40Ω resistance shows full range deflection for a current of 4mA. How can this galvanometer be converted into a voltmeter of range 0-12V?

Solution:

As we know that $V = I_G (R_G + R)$

$R = V/ I_G - R_G$

$= (12/ (4\times10^{-3})) - 40$

R = 2960 Ω

2.8 Advantages and Disadvantages of a Moving Coil Galvanometer

Advantages

- High sensitivity.
- Not easily affected by stray magnetic fields.
- The torque to weight ratio is high.
- High accuracy and reliability.

Disadvantages

- It can be used only to measure direct currents.
- Develops errors due to factors like aging of the instrument, permanent magnets and damage of spring due to mechanical stress.

2.9 Moving Iron Instruments

The moving iron instruments or MI instruments are quite cheap in cost, simple to construct and very accurate at fixed power supply frequency. These instruments can be used on AC as well as DC. Therefore, they are widely used in laboratories and on switching panels. These instruments are used either as ammeters or voltmeters only. Moving iron instruments are of two types, namely;

- Attraction type moving iron instruments.
- Repulsion type moving iron instruments.

2.9.1 Attraction Type Moving Iron Instrument

When a soft iron piece (vane) is placed in the magnetic field of a current-carrying coil, then the piece is attracted towards the center of the coil. This is because the piece tries to occupy a position of minimum reluctance. Thus, a force is exerted on the soft iron piece and deflection in the needle takes place. A moving iron instruments consists of a stationary hollow cylindrical coil. An oval shaped soft iron piece is mounted eccentrically to the spindle to which a pointer (needle) is attached. The controlling torque is provided by gravity control method while damping torque is provided by air friction as shown in the figure 2.3.

Working

When a moving iron instrument is connected in a circuit, the operating current flows through its stationary coil. A magnetic field is set up and the soft iron piece is magnetized which is attracted towards the center of the coil. Thus the pointer attached to the spindle is deflected over the calibrated scale.

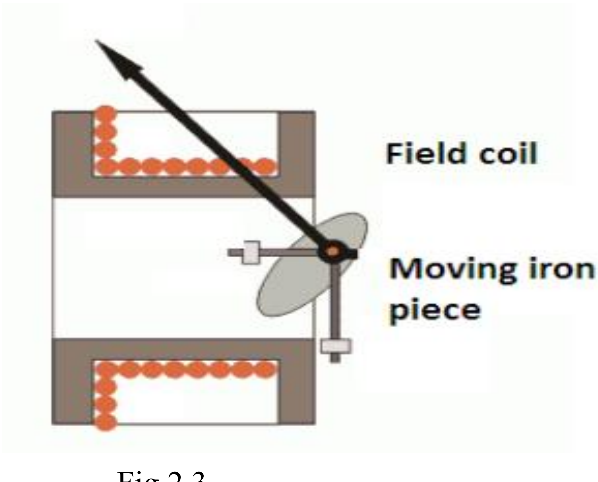


Fig 2.3

If the current in the coil is reversed the direction of magnetic field produced by the coil will reverse, also the magnetism produced in the soft, iron piece will reverse. Hence, the direction of deflecting torque remains unchanged**.** Thus attraction type moving iron instruments can be used on AC as well as on DC systems.

Since deflection θ is proportional to the square of current flowing through the coil, therefore, the scale of such an instrument is non-uniform, being crowded in the beginning. However, by choosing proper dimensions, shape and position of the soft iron piece (vane), it is possible to design and construct an instrument with a scale which is very nearly uniform over a considerable part of its length.

2.9.2 Repulsion Type Moving Iron Instrument.

Figure 2.4 shows the construction and component parts of repulsion type moving iron instrument The basic principle of a repulsion type moving iron instruments is that the repulsive forces will act when two similarly magnetized iron pieces, are placed near each other. It consists of a fixed cylindrical hollow coil which carries the operating current. Inside the coil, there are two soft iron pieces (rods or vanes) placed parallel to each other and along the axis, of the coil.One of the rod or vane is fixed and the other is movable connected to the spindle. A pointer is attached to the spindle which gives deflection on the scale. The controlling torque is provided by spring control method while damping torque is provided by air friction.

Working

When the instrument is connected in the circuit, the operating current flows through the stationary (or fixed) coil. A magnetic field is setup along the axis of the coil, this field magnetizes both the pieces similarly i.e. both the pieces attain similar polarities. A force of repulsion acts between the two, therefore movable piece moves away from the fixed piece. Thus the pointer attached to the spindle deflects over the calibrated scale.

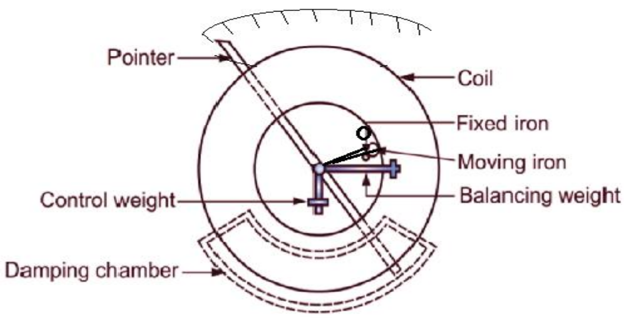


Fig 2.4

If the current in the coil is reversed, the direction of magnetic field produced by the coil is reversed. Though the polarity of the magnetized soft iron pieces is reversed but sill they are magnetized similarly and repel each other. Hence, the direction of deflecting torque remains unchanged. Thus, repulsion type moving iron instrument can be used on DC as well as on AC systems.

Since deflection θ is proportional to the square of current flowing through the coil, therefore, the scale of repulsion type moving iron instruments is non-uniform, being crowded in the beginning. However, by using tongue-shaped iron pieces, the scale of such instruments can be made uniform.

2.9.3 Advantages of Moving Iron Instruments

- The moving iron instruments are cheap, robust and simple in construction.

- These instruments can be used on both AC and DC.
- These instruments are reasonably accurate.
- The moving iron instruments possess high operating torque.
- These instruments can withstand overloads momentarily.

2.9.4 Disadvantages of Moving Iron Instruments

- The scale of moving iron instrument is non-uniform; crowded at the beginning, therefore, accurate readings are not possible at this end.
- These instruments are not very sensitive.
- Power consumption is quite high in the moving iron instruments.
- Errors are introduced due to change in frequency in case of AC measurements.

2.9.5 Errors in Moving Iron Instruments

There are two types of errors which occur in moving iron instruments:

- Errors with both AC and DC.
- Errors with AC only.

Errors with both AC and DC

The following are the main errors in moving iron instruments**,** when these are used either on DC or AC.

Error due to hysteresis: Because of hysteresis in the iron parts of the operating system, the readings are higher for descending values but lower for ascending values. The hysteresis error is considerably reduced by using mumetal or permalloy which have negligible hysteresis loss.

Error due to stray magnetic fields: Since the operating magnetic field of the moving iron instruments in comparatively weak, therefore, stray fields (fields other than the operating magnetic field) affect these instruments considerably. Thus the stray fields cause serious errors. These errors can be minimized by using an iron case or a thin iron shield over the working parts.

Error due to temperature: The effect of temperature change on moving iron instrument arises mainly from the temperature coefficient of spring. With the change in temperature stiffness of the spring varies which causes errors. However, for voltmeters, both the temperature coefficient of spring and temperature coefficient of resistance of voltmeter circuit may balance each other.

Errors with AC only

Error in moving iron instruments due to change in frequency: The change in frequency produces a change in impedance of the coil and change in magnitude of eddy currents. The increase in impedance of the coil with the increase in frequency causes serious errors in case of voltmeters only.

However, this error can be eliminated by connecting a condenser of suitable value in parallel with the swamp resistance `r' of the instrument. The impedance of the whole circuit of the instrument becomes independent of frequency if $C = L/r^2$, where C is the capacitance of the condenser.

A moving iron instrument can be used for ac or dc. But they are mainly used for AC measurements viz. AC currents and voltages. They also can measure DC quantities but their characteristics are inferior to that of permanent moving coil instruments. Therefore, in DC measurements permanent moving coil instruments are preferred and used.

CHAPTER THREE

3.0 Multimeter and Oscilloscope

3.1 Multimeter

A multimeter, also known as a volt-ohm meter, is a handheld instrument used to measure electrical voltage, current (amperage), resistance, and other values. Multimeters come in analog and digital versions and are useful for everything from simple tests, like measuring battery voltage, to detecting faults and complex diagnostics. They are one of the tools preferred by electricians for troubleshooting electrical problems on motors, appliances, circuits, power supplies, and wiring systems.

3.1.1Analog Multimeters

An analog multimeter is based on a microammeter (a device that measures amperage, or current) and has a needle that moves over a graduated scale. Analog multimeters are less expensive than their digital counterparts but can be difficult for some users to read accurately. Also, they must be handled carefully and can be damaged if they are dropped.

Analog multimeters typically are not as accurate as digital meters when used as a voltmeter. However, analog multimeters are useful for detecting slow voltage changes because you can watch the needle moving over the scale. Analog testers are exceptional when set as ammeters, due to their low resistance and high sensitivity, with scales down to 50µA (50 microamperes).

3.1.2 Digital Multimeters

Digital multimeters are the most commonly available type and include simple versions as well as advanced designs for electronics engineers. In place of the moving needle and scale found on analog meters, digital meters provide readings on an LCD screen. They tend to cost more than analog multimeters, but the price difference is minimal among basic versions. Advanced testers are much more expensive. Digital multimeters typically are better than analog in the voltmeter function, due to the higher resistance of digital. But for most users, the primary advantage of digital testers is the easy-to-read and highly accurate digital readout.

Fig 3.1

3.1.3 How to Use Multimeter

The basic functions and operations of a multimeter are similar for both digital and analog testers. The tester has two leads—red and black—and three ports. The black lead plugs into the "common" port. The red lead plugs into either of the other ports, depending on the desired function. After plugging in the leads, you turn the knob in the center of the tester to select the function and appropriate range for the specific test. For example, when the knob is set to

"20V DC," the tester will detect DC (direct current) voltage up to 20 volts. To measure smaller voltages, set the knob to the smaller ranges.

To take a reading, place the bare metal pointed end of each lead to one of the terminals or wires to be tested. The voltage (or other value) will read out on the tester. Multimeters are safe to use on energized circuits and equipment, provided the voltage or current does not exceed the maximum rating of the tester. Also, you must be careful never to touch the bare metal ends of the tester leads during an energized test because you can receive an electrical shock.

3.1.4 Multimeter Functions

Multimeters are capable of many different readings, depending on the model. Basic testers measure voltage, current, and resistance. It can also be used to check continuity, a simple test to verify a complete circuit. More advanced multimeters can be used test for all of the following values:

- AC (alternating current) voltage and current
- DC (direct current) voltage and current
- Resistance (ohms)
- Capacitance (farads)
- Conductance (siemens)
- Decibels
- Duty cycle
- Frequency (Hz)
- Inductance (henrys)
- Temperature Celsius or Fahrenheit

Accessories or special sensors can be attached to some multimeters for additional readings, such as:

- Light level
- Thermocouple
- Acidity
- Alkalinity
- Wind speed
- Relative humidity

3.2 Oscilloscope

An oscilloscope is a laboratory instrument commonly used to display and analyze the waveform of electronic signals. In effect, the device draws a graph of the instantaneous signal voltage as a function of time.

A typical oscilloscope can display alternating current (AC) or pulsating direct current (DC) waveforms having a frequency as low as approximately 1 hertz (Hz) or as high as several megahertz (MHz). High-end oscilloscopes can display signals having frequencies up to several hundred gigahertz (Ghz). The display is broken up into so-called horizontal divisions (hor div) and vertical divisions (vert div). Time is displayed from left to right on the horizontal scale. Instantaneous voltage appears on the vertical scale, with positive values going upward and negative values going downward.

The oldest form of oscilloscope, still used in some labs today, is known as the cathode-ray oscilloscope. It produces an image by causing a focused electron beam to travel, or sweep, in patterns across the face of a cathode ray tube (CRT). More modern oscilloscopes electronically replicate the action of the CRT using a liquid crystal display (liquid crystal display) similar to those found on notebook computers. The most sophisticated oscilloscopes employ computers to process and display waveforms. These computers can use any type of display, including CRT, LCD, and gas plasma.

In any oscilloscope, the horizontal sweep is measured in seconds per division (s/div), milliseconds per division (ms/div), microseconds per division (s/div), or nanoseconds per division (ns/div). The vertical deflection is measured in volts per division (V/div), millivolts per division (mV/div), or microvolts per division (?V/div). Virtually all oscilloscopes have adjustable horizontal sweep and vertical deflection settings.

The illustration shows two common waveforms as they might appear when displayed on an oscilloscope screen. The signal on the top is a sine wave; the signal on the bottom is a ramp wave. It is apparent from this display that both signals have the same, or nearly the same, frequency. They also have approximately the same peak-to-peak amplitude. Suppose the horizontal sweep rate in this instance is 1 μs/div. Then these waves both complete a full cycle every 2 μs, so their frequencies are both approximately 0.5 MHz or 500 kilohertz (kHz). If the vertical deflection is set for, say, 0.5 mV/div, then these waves both have peak-to-peak amplitudes of approximately 2 mV.

These days, typical high-end oscilloscopes are digital devices. They connect to personal computers and use their displays. Although these machines no longer employ scanning electron beams to generate images of waveforms in the manner of the old cathode-ray "scope," the basic principle is the same. Software controls the sweep rate, vertical deflection, and a host of other features which can include:

Storage of waveforms for future reference and comparison

Display of several waveforms simultaneously

Spectral analysis

Portability

Battery power option

Usability with all popular operating platforms

Zoom-in and zoom-out

Multi-color displays

The block diagram of athode ray oscilloscope is shown in the figure 3.2 below.

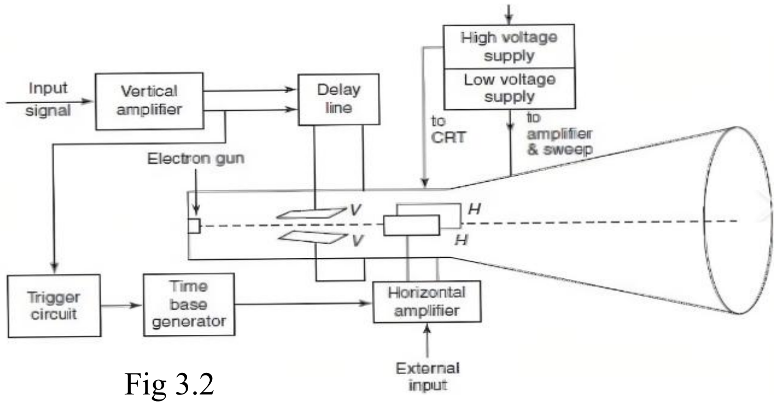


Fig 3.2

3.3 Measurement of Current, Voltage and Resistance

3.3.1 Current

Current is the measure of the rate of flow of electric charges across the conductor. It is measured in the unit of Ampere. This current measurement in a circuit is mostly done by Ammeter. Ammeter measures the electric current in the circuit. The name is derived from the SI unit of electric current, ampere. To measure electric current in a circuit, ammeter must be connected in series because, in series connection, ammeter experiences the same amount of current that flows in the circuit. Ammeter is designed to work with a small fraction of volt. So voltage drop must be minimal.

Fig 3.3 Symbol of Ammeter

The capital **A** represents the ammeter in the circuit as shown in figure 3.3

3.3.2 Voltage

Voltmeter measures the potential difference/voltage between two points in a circuit. Voltage is the difference of electrical potential between two points of an electrical or electronic circuit. It is measured in Volts. To measure p.d/voltage in a circuit, voltmeter must be connected in parallel.

The symbol is shown in figure 3.4 below

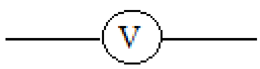


Fig 3.4 Symbol of Voltmeter

Here is the step by step procedure to be followed by users who are using a voltmeter:

1. Plug the throngs of the voltmeter into the circuit or meter, whose voltage you would like to check or measure. The red color probe is positive while the black color is negative.
2. Turn ON the selector dial or switch to the type of measurement you want to measure. For measuring direct current, i.e., in a battery, use DCV. To measure the AC, i.e., alternating current in the wall outlet, use ACV.
3. Select the settings of the range. The dial will be having options ranging from 5 to 1000. These options you can see on the DCV end and 10 to 1000 on the ACV end. You need to set the top end of the voltage you want to read. However, it is important to note that not all voltmeters will be having the same kind of settings.
4. Now turn ON the meter.
5. You need to hold the probes, i.e. positive and negative probes on both sides and touch the red side of the probe to the DC circuit or to the AC circuit and the black probe to the other side of either of these circuits.
6. Read the voltage that is displayed either on the analog or digital signal.

3.3.3 Resistance Measurement Using Ohmmeter

Resistance values of materials are measured in Ohms. It is a measure of limitation a body offers to the flow of electric current. Large values are measured in kilo-ohms and mega-ohms.

Here are steps on how to use ohmmeter to measure the resistance of a resistor or some components.

The first important step to do is turning off the circuit (To ensure the accuracy in measurement and not to damage the circuit, disconnect the power to the circuit)

Place the probes on the respective sockets:

Insert the black probe to the common (COM) port of the multimeter

Insert the red probe to the voltage (VmAΩ) port of the multimeter (Red terminal)

Zero the meter by placing the two probes touching together and adjust the dial or knob by turning it until it shows zero ohms

Test the resistance of the component using the two probes by placing at each side

Turn the knob to the relevant ohms range you are trying to measure. (If the pointer deflects to the right side of the scale, decrease the range to one level. If the pointer deflects to the left side of the scale, increase the range to one level)

Take the reading on the scale and multiply it with the appropriate range on the dial.

Make the ohmmeter zero again before testing another component.

Apart from the highlighted steps above, resistance of a body/resistor can also be obtained using Ohm’s law, colour codes, metre-bridge, etc.

3.4 Measurement of frequency, amplitude and phase

3.4.1 Measurement of Frequency

Frequency describes the number of waves that pass a fixed point in one second or the number of cycle a wave completes in one second. It is measured in Hertz.

•A simple method of determining the frequency of a signal is to estimate its periodic time from the trace on the screen of a CRT.

•To calculate the frequency of the observed signal, one has to measure the period, i.e. the time taken for 1 complete cycle, using the calibrated sweep scale. The period could be calculated by

•T = (no. of squares in cm) x (selected Time/cm scale)

•Once the period T is known, the frequency is obtained by f (Hz)= 1/T(sec)

3.4.2 Measurement of Phase

Phase is the term used to express the position of of a wave or signal at a particular point in time. It is measured as an angle in degree or radian. The value depends on what point along x-axis and at what time it is observed.

The calibrated time scales can be used to calculate the phase shift between two sinusoidal signals of the same frequency. If a dual trace or beam CRO is available to display the two signals simultaneously (one of the signals is used for synchronization), both of the signals will appear in proper time perspective and the amount of time difference between the waveforms can be measured.

This, in turn can be utilized to calculate the phase angle between the two signals ∅, between the two signals.

$$\emptyset = \frac{phase\ shift\ in\ cm}{one\ period\ in\ cm} X360 \tag{3.1}$$

CHAPTER FOUR

4.0 THE THERMOCOUPLE AND POTENTIOMETER

4.1 THE THERMOCOUPLE

Thomas Seebeck made a discovery in 1821 that when two wires of dissimilar metals are joined at both ends and one of the ends is heated such that there is temperature difference between the two ends, a continuous current flows in the thermoelectric circuit. This phenomenon is known as the Seebeck effect and is the basis for all thermocouples. All dissimilar metals exhibit this effect. For small changes in temperature Seebeck voltage is linearly proportional to difference in temperature:

$$\Delta e_{AB} = \alpha \Delta T$$

Where α is the Seebeck coefficient which is the constant of proportionality

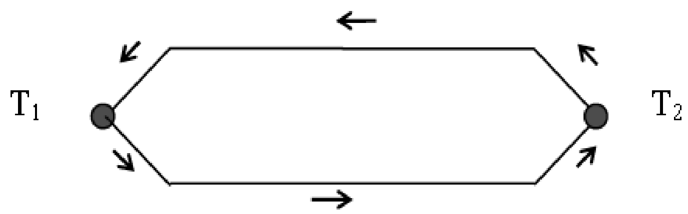


Fig 4.1

A thermocouple is a type of temperature sensor, which is made by joining two dissimilar metals at one end. The joined end is referred to as the hot junction while the other end of these dissimilar metals is referred to as the cold end or cold junction. The cold junction is actually formed at the last point of thermocouple material.

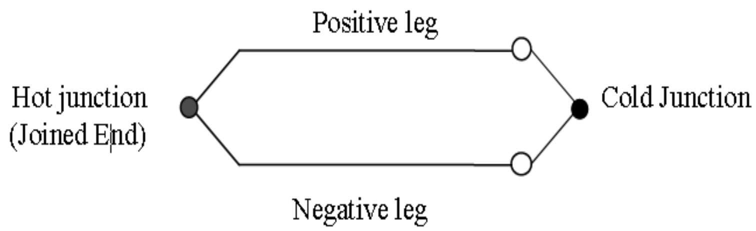


Fig 4.1

Certain combinations of metals must be used to make up the thermocouple pairs.
If there is a difference in temperature between the hot junction and cold junction, a small voltage is created. This voltage is referred to as an EMF (electro-motive force) and can be measured and in turn used to indicate temperature. The voltage created by a thermocouple is extremely small and is measured in terms of millivolts (one millivolt is equal to one thousandth of a volt). In fact, the human body creates a larger millivolt signal than a thermocouple. To establish a means to measure temperature with thermocouples, a standard scale of millivolt outputs was established. This scale was established using 32 deg. F (0°C) as the standard cold junction temperature (32 deg. F (0°C) = 0 millivolts output).
Thermocouples can measure temperature ranges from -200°C to 1300°C. The table below shows different types of thermocouple and their constituent metals.

Table 4.1. Thermocouples and their Constituent Metals

Thermocouple Type	Composition	Temperature Range(°C)	Colour
B	Platinum 30%, Rhodium	1370 - 1700	(+) Grey (-) White
C	Tungsten 5%, Rhenium	1650 - 2315	
E	Chromel, Constantan	95 - 900	(+) Purple (-)White
J	Iron and Constantan	95 - 760	(+) Black (-)White
K	Chromel and Alumel	95 - 1260	(+)Green (-)White
M	Nickel and Nickel	0 - 1287	
N	Nicrosil and Nisil	650 - 1260	(+)Pink (-)White
R	Platinum 13%, Rhodium	870 - 1450	(+)Orange (-)White
S	Platinum 10%, Rhodium	980 - 1450	(+)Orange (-)White
T	Copper, Constantan	-200 -350	(+)Brown (-)White

Factors to be Considered while Choosing a Thermocouple

1. Temperature range
2. Thermal conduction requirement
3. Thermocouple availability
4. Installation requirement
5. Cost

4.2 Potentiometer

Potentiometers are made of resistive materials. These materials include carbon composition, Wire-wound, conductive plastic, and cermet. A potentiometer is sometimes called **POT** which is short for potentiometer. It is a type of variable resistors. A potentiometer can therefore be considered as a three-terminal variable resistor that can be adjusted manually to control the flow of current in a circuit. A potentiometer can either be a linear or rotary type. The symbol of potentiometer is shown below.

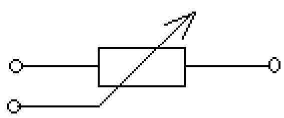


Fig 4.3 Symbol of Potentiometer

4.2.1 Rotary Potentiometer

It consists of a wiper which moves on the semi-circular resistance material that is placed between the potentiometer's two terminal contacts with the help of a knob. It is normally used for obtaining an adjustable supply voltage to an electrical circuit or a part of an electronic circuit. The radio transistor's volume controller has a rotary potentiometer and its knob is used to control the supply to the amplifier.

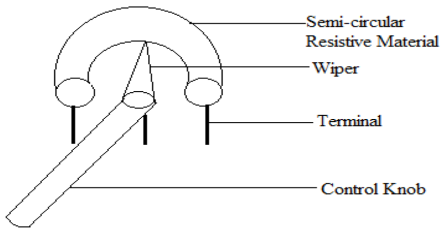


Fig 4.4

4.2.2 Linear Potentiometer

It works the same as a rotary potentiometer but the movement of the wiper is linear on the straight resistive material. Linear potentiometers can be used to measure the battery cell's internal resistance, the voltage across a circuit's branch, etc. the most popular example of using such potentiometers is in the equalizer of music and sound mixing systems.

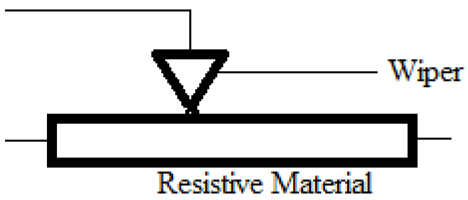


Fig 4.5

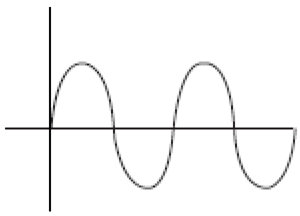
4.2.3 Applications of Potentiometer

(i) It is used as voltage divider
(ii) It is used as transducer for measuring short distance.
(iii) It can be used for measuring/comparing the e.m.f. of cells
(iv) Used in audio mixer and volume control in radio and television system.
(v) It is used for frequency attenuation.

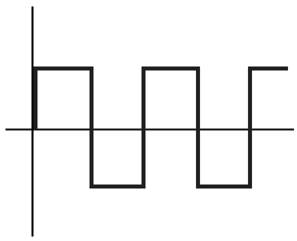
CHAPTER FIVE

5.0 SIGNAL GENERATOR

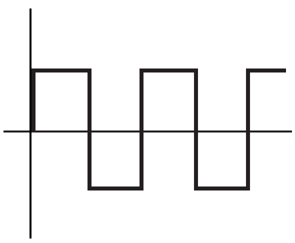
A signal generator is an instrument that provides a controlled waveform or signal for use in testing, aligning or measurement on other circuit or equipment. Examples of waveform produced by signal generator are sine wave, square wave, triangular wave, e.t.c. It produces electrical signal in form of a wave. It delivers an accurately calibrated signal at different frequencies. Different types of waveforms produced by signal generators are pictorially represented below.



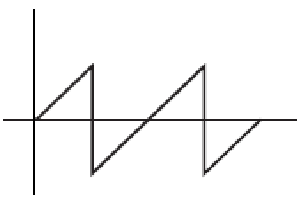
(a) Sinewave



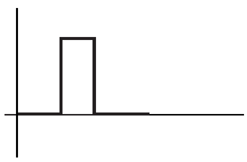
(b) Square wave



(b) Square wave



(c) Saw-tooth wave



(d) Pulse

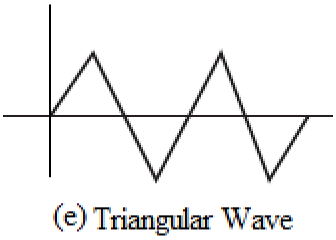


Fig 5.1

5.1 Types of Signal Generators

They can be classified into audio signal generator, function generator, pulse generator, RF generator, frequency synthesizer and RF marker.

5.2 Audio Signal Generator

As the name implies this type of signal generator is used for audio applications. Signal generators such as these run over the audio range, typically from about 20 Hz to 20 kHz and more, and are often used as sine or square wave generators. They are often used in audio measurements of frequency response and for distortion measurements. As a result they must have a very flat response and also very low levels of harmonic distortion. The block diagram of audio generator is shown below.

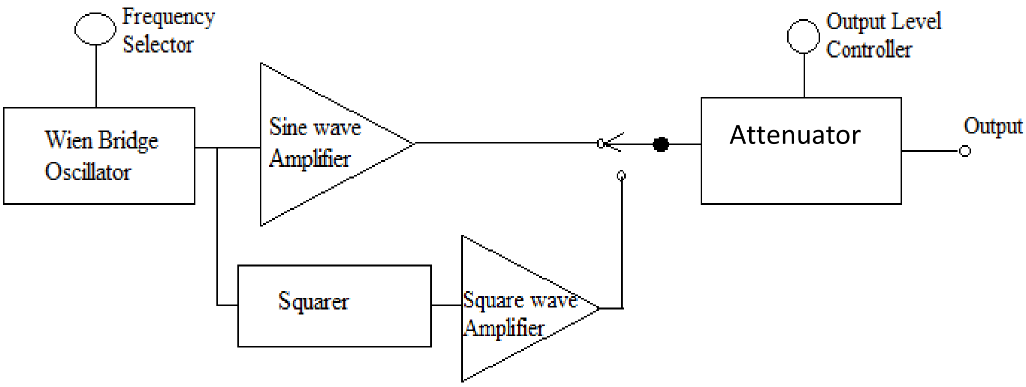


Fig 5.2

Applications

(i) It is used for testing the frequency response of audio amplifier
(ii) It is used together with oscilloscope to determine the frequency of a signal.

(iii) For distortion measurement

5.3 Function Generator

The function generator is a type of signal generator that is used to generate simple repetitive waveforms. Typically this signal generator will produce waveforms or signals such as sine waves, saw-tooth waveforms, square and triangular waveforms. Early function generators tended to rely on analogue oscillator circuits that produced the waveforms directly. Modern function generators may use digital signal processing techniques to generate the waveforms digitally and then convert them from the digital into an analogue format. Many function generators will tend to be limited to lower frequencies as this is where the waveforms created by this type of signal generator are often required. However it is possible for higher frequency versions to be obtained. They produce signals with frequencies ranging from 0.1 Hz to 3 mHz. The block diagram of function generator is shown below.

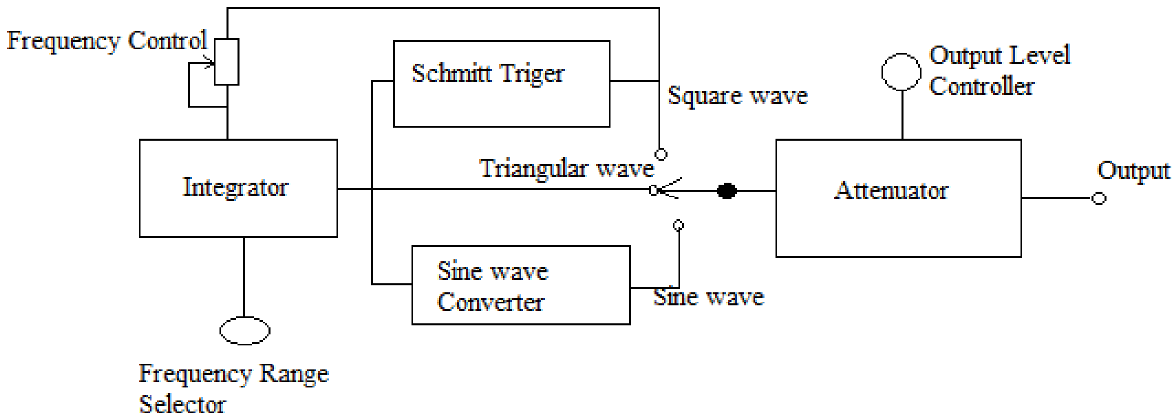


Fig 5.3

Applications

(i) To generate signals to test both analog and digital systems
(ii) For repair
(iii) For calibration
(iv) Used in design process
(v) Used for testing amplifiers, filters

5.4 Pulse Generator

As the name suggests, the pulse generator is a form of signal generator that creates pulses. These signal generators are often in the form of logic pulse generators that can produce pulses with variable delays and some even offer variable rise and fall times. Pulses are often needed when testing various digital circuits, and sometimes analogue circuits. The ability to generate pulses enables circuits to be triggered, or pulses trains to be sent to a device to provide the required stimulus.

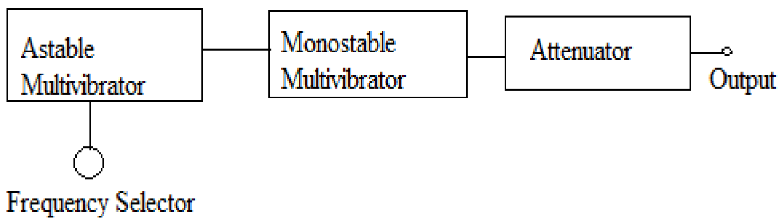


Fig 5.4

Applications

It is used to;

(i) test memory circuit
(ii) test shift registers
(iii) Counters
(iv) Digital systems

5.5 RF signal generator

As the name indicates, this type of signal generator is used to generate RF or radio frequency signals. Typical RF radio frequency signal generator A typical RF signal generator. An RF signal generator may use a variety of methods to generate the signal. Analogue signal generator types used free running oscillators, although some used frequency locked loop techniques to improve stability. However most RF signal generators use frequency synthesizers to provide the stability and accuracy needed. Both phase locked loop and direct digital synthesis techniques may be used. The RF signal generators often have the capability to add modulation to the waveform. Lower end ones may be able to add AM or FM, but high end RF signal generators may be able to add modulation formats OFDM, CDMA, etc. so they can be used for testing cellular and wireless systems. It produces signal in range of 100 kHz to 40 GHz. Each section of RF generator shown below is shielded to prevent frequency interference between the components and to prevent emission of RF energy from any section except the output terminal. The block diagram of the generator is shown below.

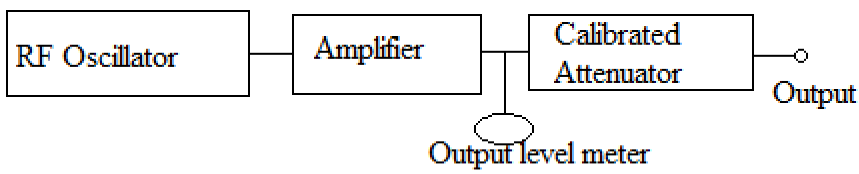


Fig 5.5

Application

(i) To perform different tests on radio transmitter and receiver

(ii) To test the amplitude and frequency response of RF amplifier
(iii) It is used for research purposes

5.6 Frequency Synthesizer

This is a type of RF generator with output frequency ranging from 1 MHz to 160 MHz. It has the same applications as RF generator.

CHAPTER SIX

6.0 PRESSURE

The amount of force exerted (thrust) on a surface per unit area is called pressure. It can be defined as the ratio of the force to the area over which the force acts.

Pressure (P) = Force/ Area

The SI unit of pressure is **'pascals (Pa)'.** 1 Pa = $1N/m^2$. It can also be measured in other units such as mmHg, torr, bar, atmosphere. It can be measured using barometers, manometers, pressure gauges, etc.

6.1 The Barometer

The barometer is a device meant for measuring the local atmospheric pressure. The figure below shows a mercury barometer which consists of a 1metre long glass tube closed at one end and completely filled with mercury and kept inverted in a bowl of mercury. A small quantity of mercury will drop into the bowl and thus a vacuum forms at the upper end of the tube.

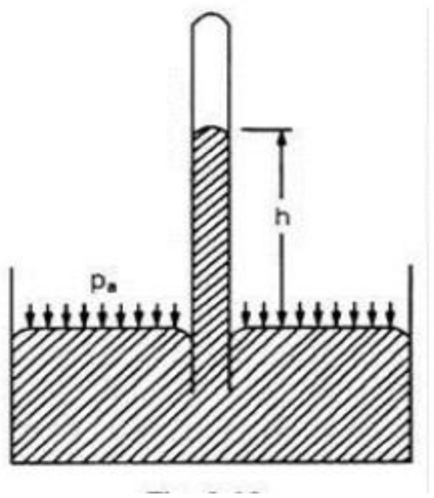


Fig 6.1

The atmospheric pressure acting on the surface of mercury in the bowl will support a mercury column in the tube. Let h be the height of mercury column in the tube measured above the surface of mercury in the bowl. The height of the mercury column at sea level, is approximately 760 mm of mercury. The space above the mercury in the tube will contain vapours of mercury. This space is called Torricellian vacuum

6.2 Aneroid Barometer:

This device consists of a partially evacuated corrugated box spring which is made from an alloy of copper and beryllium. It is prevented from collapsing by a strong spring. Pressure variations cause the front part of the box to deform inwards or outwards so that the pull of the spring will just resist the force due to the pressure of the atmosphere. These small displacements are amplified and move a pointer provided over a calibrated scale.

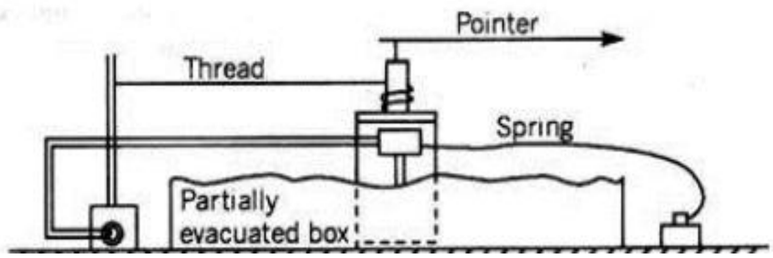


Fig 6.2

Any change in the external pressure would cause this box to contract or expand. A mechanical system is installed to amplify and indicate these changes. This is displayed using a needle and a dial. Aneroid barometers can be easily calibrated. The sensitivity of the aneroid barometer can be changed by changing the cell or the amplifying system.

Difference Between Aneroid and Mercury Barometer

(i) Aneroid barometer is a type of barometer that uses a small flexible metal box to measure the atmospheric pressure while mercury barometer is a simple barometer that uses mercury to measure the atmospheric pressure.

(ii) Aneroid barometer uses a metal box to measure the atmospheric pressure but mercury barometer uses mercury to measure the atmospheric pressure.

(iii) Aneroid barometer measures the external pressure using the expansion or contraction of a flexible metal box which leads to moving a needle on a pressure scale to get the reading whereas mercury barometer measures the external pressure by measuring the rise or the height of mercury inside the vertical glass tube.

(iv) Since an aneroid barometer is a solid and compact device, it is easy to handle and transport but mercury barometer is large (about 3 feet high) and fragile it is hard to handle and transport.

(v) It is easy to take a measurement from the aneroid barometer since it directly gives a value but it is hard to take a measurement from the mercury barometer since the height should be measured accurately after it gets balanced.

(vi) The aneroid barometer need machinery for its construction while mercury barometer needs no machinery for the construction.

(vii) Aneroid barometers are more stable than mercury barometers

(viii) Aneroid barometers are used in homes, recreational boats, aircrafts, etc. while mercury barometers are mainly used in laboratories, for weather forecasting, to determine the altitude of a place, etc.

6.3 Manometer

A manometer is a scientific instrument used to measure gas pressures. Open manometers measure gas pressure relative to atmospheric pressure. A mercury or oil manometer measures gas pressure as the height of a fluid column of mercury or oil that the gas sample supports. In the manometer, one end connects to a gas-tight seal to test the pressure source. The other end of the tube is left open to the atmosphere and it will be subjected to the pressure of approximately 1 atmosphere (atm). If the test pressure is greater than the pressure of 1 atm (atmosphere), the liquid in the column will be forced down by the pressure and consequently cause the liquid of the reference column to rise by an equal amount. The pressure applied by a column of liquid is given by the equation $P = \rho hg$. In this equation, P is the calculated pressure, h is the fluids heights, g is the gravitational force and ρ is the density of the liquid. Also, it is measuring a differential pressure rather than absolute pressure. Furthermore, we use the substitution $P = Pa - Po$. Besides, in this situation Pa is the test pressure and P0 is the reference pressure.

6.4 Bourdon Gauge

This device consists of a metallic tube of elliptical section closed at one end A, the other end B being fitted to the gauge point where the pressure is to be measured. As the fluid enters the tube, the tube tends to straighten. By using a pinion-sector arrangement the small elastic deformation of the tube is communicated to a pointer in an amplified manner. The pointer moves over a graduated dial. The device is calibrated by subjecting it to various known pressures. The Bourdon gauge is suitable for measuring not only high pressures such as those in a steam boiler or a water main but also negative or vacuum pressures. A gauge which is so devised to measure positive as well as negative pressures is called a compound gauge.

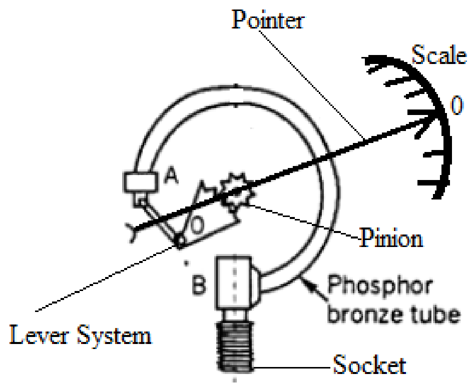


Fig 6.3

CHAPTER SEVEN

7.0 Recorder and Reproducer

They are electrical, mechanical, electronic or digital devices for inscription and recreation of signals such as sound and visual signals. The two main classes of sound recording and reproducing technology are analog and digital.

7.1 Analog Recording

This is achieved by microphone diaphragm that senses changes in atmospheric pressure caused by sound waves and record them as mechanical representation of sound wave on a medium such as phonographs. Sound waves can also be recorded using magnetic tape in which sound wave is picked up by microphone and are converted into varying electric current which is transformed into varying magnetic field by an electromagnet which makes a representation of the sound as magnetized area on a plastic tape with magnetic coating on it. Meanwhile analog signal reproduction is the reverse process with a bigger loud speaker diaphragm causing changes to atmospheric pressure to form acoustic sound wave.

7.2 Digital Recording

This involves converting analog sound signal picked up by the microphone to a digital form by the process of sampling. This allows the audio data to be stored or transmitted by different media. Digital recorders store audio signal as a series of binary numbers representing samples of the amplitude of audio signal at equal time intervals, at sampling rate high enough to convey all sounds that can be heard. This digital audio signal must be reconverted to analog form during playback before being amplified and connected to a loud speaker to produce sound.

7.3 Types of Recorders

There are different types of recorders. The commonest are chart, audio and video recorders.

7.3.1 Chart Recorder

This is an electromechanical device/system that records an electrical or mechanical input trend onto a piece of paper or virtually. The paper is passed under a pen and the pen is deflected in proportion to the signal. The result is a graph or chart of the data. A chart recorder can have more than one pen with different colours. The figure below shows a typical chart recorder with one pen.

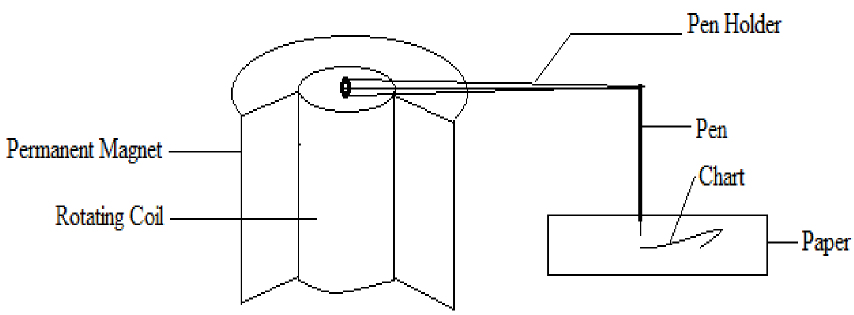


Fig 7.1

Types of Chart Recorder

When limited variables are required which do not need PC-based interface, a chart recorder can be used.Some types of chart recorder are strip chart recorder, XY recorder, hybrid recorder and paperless recorder.

1. Strip Chart Recorder

Strip Chart Recorder is the one in which data is recorded on a continuous roll of chart paper moving at a constant speed. The recorder records the variation of one or more variables with respect to time. The Strip Chart Recorder consists of a pen (stylus) used for making marks on a movable paper, a pen (stylus) driving system, a vertically moving long roll of chart paper and chart paper drive mechanism and a chart speed selector switch as shown in the figure below.

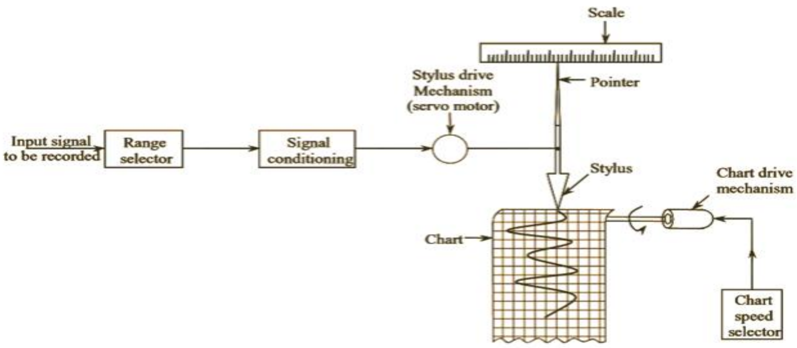


Fig 7.2

Most strip chart recorders use a servo feedback system, to ensure that the displacement of the pen (stylus) across the paper tracks the input voltage in the required frequency range. A potentiometer system is generally used to measure the position of the writing head (stylus). The chart paper drive system generally consists of a stepping motor which controls the movement of the chart paper at a uniform rate. The data on the strip chart paper can be recorded by various methods.

2. Circular Chart Recorder

Circular chart recorder is a data acquisition tool that archive data points onto a uniformly rotating circular chart over a timed interval in proportion to the signal received. Circular chart recorder provides real-time output and is ideal when a hardcopy of data is needed. Here the circular paper is spun under one or more pens and the chart is designed to rotate in a standard time period.

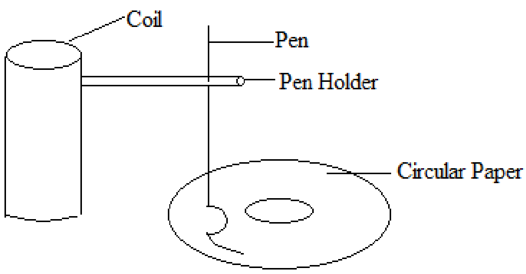


Fig 7.2

3. X-Y Recorder

The X-Y recorder is used to plot one variable against another variable. In an analog X-Y recorder, the writing head is deflected in either the x-direction or the y-direction on a fixed graph chart paper. The graph paper used is generally squared shaped, and is held fixed by electrostatic attraction or by vacuum. The writing head is controlled by a servo feedback system or by a self-balancing potentiometer. The writing head consist of one or two pens, depending on the application. In practice, one e.m.f is plotted as a function of another e.m.f in an X-Y recorder.

In some cases, the X-Y recorder is also used to plot one physical quantity (displacement, force, strain, pressure, etc.) as a function of another physical quantity, by using an appropriate transducer, which produces an output (EMF) proportional to the physical quantity. The motion of the recording pen in both the axis is driven by servo-system, with reference to a stationary chart paper. The movement in x and y directions is obtained through a sliding pen and moving arm arrangement.

Applications of X-Y Recorders

These recorders are used to measure the following.

1. Speed-torque characteristics of motors.
2. Regulation curves of power supply.
3. Plotting characteristics of active devices such as vacuum tubes, transistors, zener diode, rectifier diodes, etc.
4. Plotting stress-strain curves, hysteresis curves, etc.
5. Electrical characteristics of materials, such as resistance versus temperature.

4. Hybrid Recorder

This can function as a recorder or data logger. It generates chart of input signals like other chart recorders. They are available in multichannel designs. As a data logger, it can generate a log or chart of various inputs. It has print head or module handling the tracking of different channels. It is cost effective.

5. **Paperless Chart Recorders**

Paperless recorders are one of the latest types of recorders to emerge on the market. Paperless recorders display the chart on the recorders' graphic display rather than print the chart on paper. The data can normally be recorded in internal memory or to a memory card for later transfer to a computer. The major benefit of paperless recorders is conservation of paper and easy transfer to a computer.

Application of Chart Recorder

Chart recorders are a familiar sight in manufacturing plants, where they track such variables as temperature, pressure, flow, pH, and humidity. Laboratories, meanwhile, use them to monitor scientific and engineering data generated in testing, diagnostics, statistical analysis, and other work requiring a graphic record.

7.3.2 Audio Recorder

These are devices used for recording audible signals. There are several types of audio recording devices. Some of them are; Cassette recorder, minidisc recorder, DAT recorder, e.t.c.

1. Cassette Recorders

Cassette recorders make analogue recordings onto compact cassettes containing 1/8" tape running at 1.875 inches/sec. Modern recorders have an acceptable frequency response for speech recordings, but typically have a poor signal-to-noise ratio. The poor SNR means that cassette recorders are unforgiving about incorrectly set recording levels - quiet recordings are very noisy. The analogue technology brings its own problems of course: significant harmonic distortion, loss of quality with storage and the problems of recordings being affected by drop outs and media damage. Although they are good recorders for speech, they are being replaced by minidisc recorders and are increasingly hard to find in markets.

2. Minidisc Recorders

Minidisc recorders make compressed digital recordings on 2.5" magneto-optical disks. The disks are protected in a cartridge. Recordings are made by compression and decompression of the sound signal . Minidisc recorders are extraordinarily small and light. However their small size can make them difficult to operate. Minidisc recorders are designed to prevent users making perfect digital copies of copyright recordings, so one cannot gain access to the compressed bit stream directly has to decompress and recompress, and this process loses quality. Few minidsc recorders have a digital output, so to obtain a digital transfer to a computer, you will also need a minidisc deck with a digital output and a PC with a digital audio input. A recent innovation in minidisc recorders is the HiMD recorder. These are interesting in two respects: firstly they use larger capacity disks and can record an hour of audio even in an uncompressed format, and secondly they have a USB interface which can be used to upload recordings to a computer (providing they were made on that recorder using an analogue input). An example of a HiMD minidisc recorder is the Sony NH700

3. **Digital Audio Tape (DAT) Recorders**

Digital Audio Tape (DAT) recorders make digital recordings on 6mm tape cartridges, using a modified video recording mechanism with a rotating tape head. DAT recorders have been used widely in professional broadcasting because of their capability for making high quality recordings. DAT recorders usually have some kind of digital output stream capability so that recordings can be transferred digitally to a computer system. One disadvantage of DAT recorders is the complexity of the rotating head and tape transport mechanism. These devices are probably relatively fragile, and this means that DAT recorders should be transported with care and regularly maintained.

4. Solid-state Recorders

Solid-state recorders make digital recordings onto memory cards, such as compact flash cards. Usually they have an option to save as uncompressed linear pulse code modulation (PCM) or in a compressed format such as MP3. For a given size of memory card, you can record for longer time using a compressed format but at a lower quality. It is recommended that data rates of at least 192kbps are used for good quality speech recordings intended for computer analysis. Compact flash card memories are available in different sizes.

5. Hard-disk Recorders

Hard-disk recorders make compressed or uncompressed digital recordings onto high-capacity disk drives inside the recorder. Recordings must then be copied from the recorder onto permanent media, such as CD or computers. This type of recorder has a 20 or 40GB hard disk and can make real-time recordings from an external microphone using MP3 compression at 320kbps. Recordings can then be uploaded to a PC using its USB interface

7.3.3 Video Recorder

A video recorder is a device for recording visible images/signals for further analysis or future reference. A video recorder may be any of the following;

(1) Digital Video Recorder
(2) Digital Versatile Disc
(3) Video cassette Recorder
(4) Video Tape Recorder

1. Digital Video Recorder

A digital video recorder (DVR) otherwise known as personal video recorder (PVR) is an electronics device designed for recording video signals in a digital format within a mass storage device such as USB flash drive, hard disk drive or any other storage device. Compared to other video recording alternatives, a digital video recorder has many advantages such as being tapeless, faster data retrieval and higher image quality. It is mostly used in home entertainment and in surveillance/security systems.

2. Digital Versatile Disc

A digital versatile disc (DVD) is an optical disc storage medium similar to a compact disc, but with enhanced data storage capacities as well as with higher quality of video and audio formats. DVD is widely used for video formats, audio formats as well software and computer files. Digital versatile discs are also known as digital video discs. The layers in the DVD are made by polycarbonate plastic. Digital versatile discs can be categorized in different ways based on their applications. If they are used for reading only and cannot be written, then they are classified as DVD-ROM. If the DVDs can be used to record any type of data, then they are called DVD-R. If the disc can be read, written and then erased and rewritten, it is called DVD-RW. They are capable of storing more data compared to CD. This is due to smaller size of the pits and bumps and high density of the tracks in the DVDs.

3. Video Cassette Recorder

This is an electromechanical device which records and plays back analog audio and video data on a removable magnetic cassette tape. It allows for time shifting in TV stations.

4. Video Tape Recorder

This is designed to record and playback video and audio material from magnetic tape

CHAPTER EIGHT

8.0 Power Supply

Sources of electrical power include d.c. or a.c. generator/dynamo, battery and solar panel. All power supplies have a power input connection, which receives energy in the form of electric current from a source, and one or more power output or rail connections that deliver current to the load. A power supply can therefore be defined as an electrical/electronic device that supplies electric power to an electrical load. Power supplies have essential component parts found in all models with additional operations added depending on the device type. Power supplies may need to change voltage up or down, convert power to direct current, or regulate power for smoother output voltage.u

8.1 Types of Power Supply

There are two types of power supplies viz; AC and DC power supply. Based on the electrical device's electric specifications it may use AC power or DC power.

8.2 AC Power Supply

This power supply supplies a.c. voltage to a load. It utilizes either step-up or step-down transformer to change the level of the a.c. voltage. Different values of AC voltages are generated by using transformer. The transformer may have multiple secondary windings or taps, in which case the power supply unit uses switches to select the different voltage levels. Alternatively, a variable transformer can be used to continuously vary the voltages. Some variable AC supplies have meters to monitor the output voltage, current, and power.

8.3 DC Power Supply

A DC power supply is one that provides a consistent DC voltage to its load. Based on its design, a DC power supply may draw it input voltage controlled from either a DC source or from an AC source like the power mains. A dc power supply can either be regulated or unregulated power supply. A regulated power supply is the type whose output power remains almost constant when either the load or source changes. Meanwhile unregulated power supply on the other hand is the one in which the output power changes significantly whenever there is a change in the power level of either the source or load.

8.4 Unregulated Power Supply

Unlike regulated power supply, the output voltage of an unregulated power supply is not regulated meaning the output voltage changes as the load varies. It does not have voltage regulation circuit. It provides a fluctuating amount of output power. It provides a predetermined output based on the input and load voltage and even a little variation in the input directly affects the output voltage. An unregulated power supply consists of a transformer, a rectifier and a filter. These little variations in the output voltage are called ripple voltage. Electronic circuits do work properly when powered with unregulated power supply. They require constant voltage supply regardless of the variations in the input voltage or load current. To do this, a voltage stabilizing device called voltage regulator is used. The circuit diagram of an unregulated power supply is shown in figure 8.1

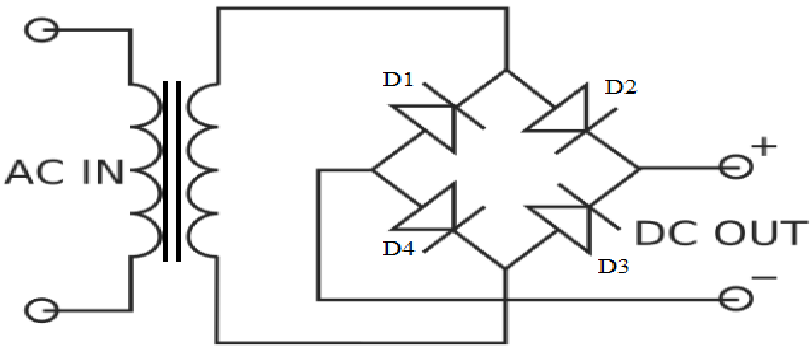


Fig 8.1

8.5 Regulated Power Supply

The regulated power supply is a kind of electronic circuit, designed to provide the stable DC voltage of fixed value across load terminals irrespective of load variations. The main function of the regulated power supply is to convert an unregulated alternating current (AC) to a steady direct current (DC). The RPS is used to confirm that if the input changes then the output will be stable. This power supply is also called a linear power supply, and this will allow an AC input as well as provides steady DC output. The block diagram of the power supply is shown in figure 8.2.

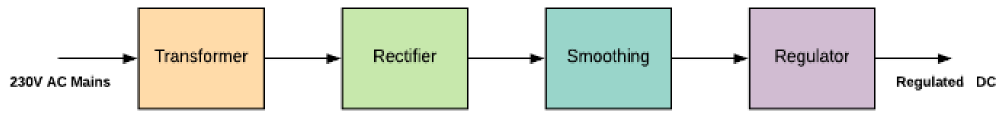


Fig 8.2

A regulated power supply is made up of transformer, rectifier, filter, regulator and load protector. A typical circuit diagram of a regulated power supply is shown in figure 8.3.

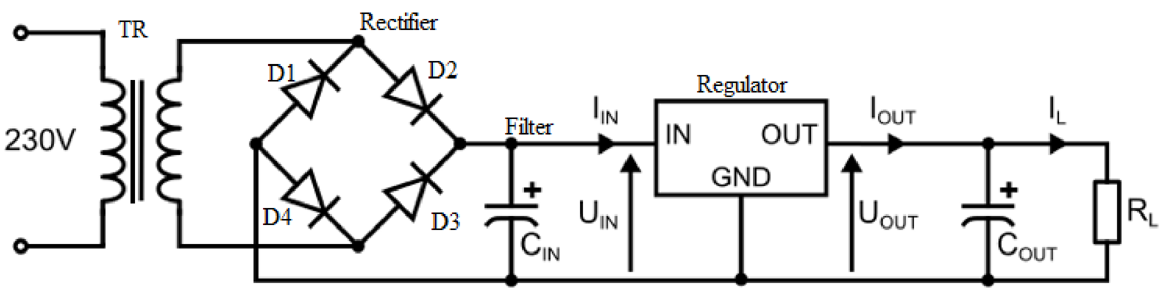


Fig 8.3

A transformer alters the AC mains voltage toward a particular value. Its function is to step up and step down the voltage. For instance, a step-down transformer is used in a transistor radio, and a step-up transformer is used in **a CRT**. Transformer gives separation from the power-line, and must be used even as any modify within voltage is not required.

Rectifier

A rectifier is an electronic circuit/device used to convert alternating current into direct current. It can be a half wave rectifier or full wave rectifier. A single silicon diode may be used to obtain a DC voltage from the AC input. The half wave rectifier conducts on only half of each cycle of the AC input wave, effectively blocking the other half cycle. The circuit diagram of half wave rectifier is shown in figure 8.4 below.

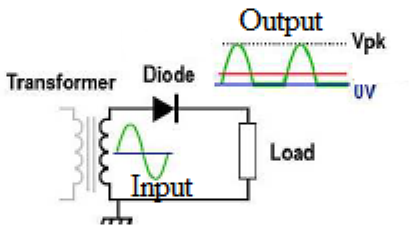


Fig 8.4

Half wave rectifier allows either positive half cycle or negative half cycle of the input AC signal and the remaining half cycle is blocked. As a result, a large power is wasted. Also, the output Direct Current (DC) produced by the half wave rectifier contains large ripples. This ripple voltage fluctuates with respect to time. So it is not suitable for practical applications. Therefore, to reduce the power loss and reduce the ripples at the output, we go for another type of rectifier known as full wave rectifier. As the name suggests, the full wave rectifier rectifies both positive and negative half cycles of the input AC signal. Even though the full wave rectifier rectify both positive and negative half cycles, the DC signal obtained at the output still contains some ripples. To reduce these ripples at the output, we use a filter. The circuit diagram of full wave rectifiers using four and two diodes are shown in figures 8.5 and 8.6 respectively.

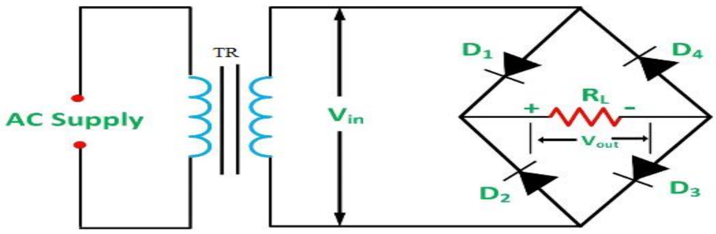


Fig 8.5

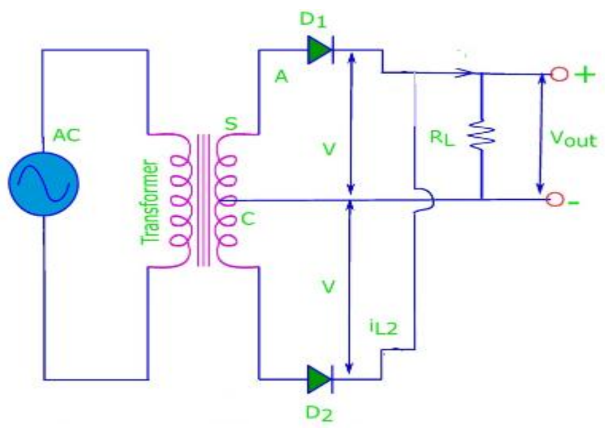


Fig 8.6

Filter

A filter in the regulated power supply is mainly used for leveling the ac differences from the corrected voltage. A filter circuit is one which removes the ac component present in the rectified output and allows the dc component to reach the load. Filters are classified into four types namely capacitor filter, Inductor filter, LC filter & RC filter. The circuit diagram of a typical filter circuit is shown below.

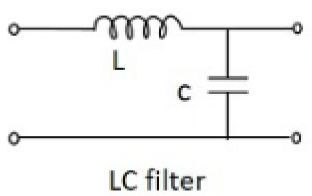


Fig 8.7

Voltage Regulator

A voltage regulator in the regulated power supply is essential for keeping a steady DC output voltage by supplying load regulation as well as line regulation. For this purpose, regulators like zener diodes, transistors, 3-terminal integrated regulators are used. An SMPS-switched mode power supply can be used for supplying huge load current by small power dissipation within the series pass transistor. A voltage regulating circuit diagram is shown below.

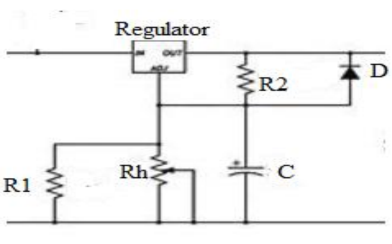


Fig 8.8

8.6 Switch Mode Power Supply (SMPS)

A switched mode power supply (also known as switching power supply) is an electronic power supply device that converts electrical power from one voltage to another efficiently.

Typically, SMPS is used to transfer power from a DC/AC source to a DC load (i.e. a computer, mobile phone, etc.). Most switched-mode power supplies convert a higher voltage (110V or 220V AC) to a much lower DC voltage like 24V, 12V, or 5V. We can find these types of power supplies in almost every electrical appliance, especially those of which are compact for example, mobile phone charging adapters, computers, laptop charging adapters, etc.

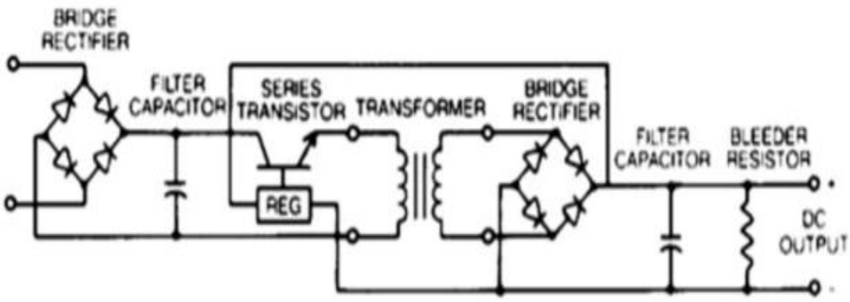


Fig 8.9

Advantages and Disadvantages of Switched Mode Power Supplies

The advantages of switched-mode power supplies are:

- They have small sizes and therefore can fit into compact devices
- Due to the semiconductor-based components, SMPS is lighter in weight
- They are more efficient than linear power supplies (70-95% typical)
- Supports wider input and output voltage ranges
- Provides additional functions like adjustable outputs and safety functions as a short circuit, over-voltage, over current, and over-temperature lockout protection
- Lower heat dissipation, thus requiring minimal active cooling

However, SMPS also have drawbacks that sometimes make them unsuitable for certain applications. For example, a SMPS is a much more complex circuit than a traditional linear circuit. Therefore, there are many components that can malfunction and hinder the performance of the power supply. Also, SMPS are known for their higher EMI (Electromagnetic Interference) and electrical noise since they work in high frequencies. A poorly designed SMPS can cause glitches and sometimes even permanently damage sensitive electronics powered by them. In the power realm, switched-mode power supplies also create harmonic distortion in the power grid and sometimes may require additional power factor correction if they are not built-in to the supply.

Uninterruptible Power Supply (UPS)

UPS is a Backup power source that, in the case of power failure or fluctuations, allows enough time for an orderly shutdown of the system or for a standby generator to start up. UPS consists usually of a bank of rechargeable batteries and power sensing and conditioning circuitry. It uses electromagnetic interference/radiofrequency interference(EMI/RFI) filter, battery and transfer switch.

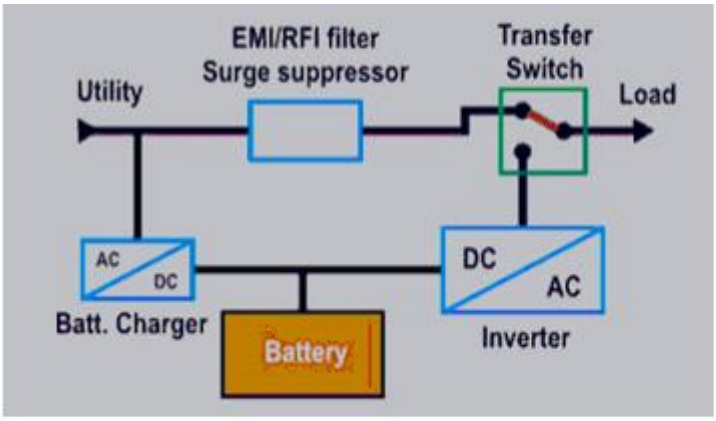


Fig 8.10

CHAPTER NINE

9.0 ELECTRONIC TROUBLESHOOTING

The unexpected behavior exhibited by a circuit is due to improper locating or soldering of components, component damage due to aging, faults, overheat, and so on. Such a type of behavior can cause undesired results or even circuit damage. Therefore, these unexpected results of the electronic circuit may require some troubleshooting and testing procedures for making it a ready to use project. Troubleshooting of an electronic circuit is a process of having a special outlook on components that comes out with remedies to repair it. Troubleshooting is also the process that determines the cause of the problem in the electronic circuit by examining the affected area of it, and by taking appropriate action.

9.1 Materials used for Troubleshooting

These materials are hand tools, power tools, and bench-mounted tools for tightening, cutting, drilling, and trimming. Other tools are used for soldering, inspecting, and finishing.

They include: - Screwdrivers

Pliers

Wire Cutters

Wire Strippers

Socket and Hex Drivers

Grinders

Drills

Hand Saws

Soldering iron

Magnifiers and Microscopes

Multimeter

Oscilloscope

Power supply

Function generators

Instruction mannuals

9.2 Troubleshooting Methods

There are basic techniques used by all troubleshooters when servicing electrical or electronic devices. It involves fault establishment, fault location and fault correction. The method can either be functional area approach method or split half method. The functional approach method involves tracing the fault to the functional unit of the system e.g amplifier, oscillator, power supply, and so on. Meanwhile the split half method is used when the faulty system comprises of many smaller units. It has to do with dividing the faulty circuit into two successive halves until fault is identified.The technique a troubleshooter uses depends on the type of defect or symptom that exists in the faulty device. Some of the troubleshooting techniques are as follows:

1. Voltage measurements
2. Current measurements
3. Resistance measurements
4. Substitution
5. Bridging
6. Heat
7. Freezing
8. Signal tracing and injection
9. Component testers, test lamps
10. Re-soldering, adjusting, etc.
11. Bypassing
12. Logic analysis and network injection

Troubleshooting steps

Step 1: Define the problem

The first step of solving any problem is to know what type of problem it is and define it well. A clear definition is fundamental when troubleshooting. Some equipment will have built-in ways of letting you know; alarms can sound, red lights flash, or a warning can go off when certain parts overheat. These signals can help with problem-solving. Other equipment just stops working. Whatever the case may be, you have to identify and define the problem before you can move forward.

Step 2: Collect relevant information

Gather all the available information about the machine and its operations from the machine manual, any data recording operations book. For example, how often the machine is used, by whom, for what, and how long. Maintenance history, problem reports, etc. are equally useful.

Step 3: Analyze collected data

Using all the information you have gathered, available checklists, and as much technical know-how as you can muster, you can now try to determine the root cause of the problem. Seek out expertise from other maintenance troubleshooters or the person who reported the fault. It is much easier to solve a problem that you have seen before.

Think about recent changes to the asset. Ask yourself:

- Did we use new replacement parts?
- Has there been an upgrade lately?
- Did we change the type of input material we use?
- Has the device been used in a different way than usual?
- Has there been an electrical surge?

Recent changes to the system or environment can often explain why the problem has come up. If you still have no clue what caused the problem after analyzing the data, you need to go back to Step 2 and collect more info. It is possible to overlook things or disregard something as unimportant during the first round of the information-gathering process. After this exercise, the person performing troubleshooting should form an educated guess and put forward some solutions.

Step 4: Propose a solution and test it

Using what you know from above, you can create your plan of attack. Get to the solution through a process of elimination and trial and error. In some cases, you may be able to test your theory on a smaller scale asset. You may have multiple options to try. Start with the simplest one first and work from there.

Take the following into account:

- potential safety concerns
- all the required resources and associated costs
- how complex the implementation will be
- the long-term outlook for the machine
- any personal biases person performing the troubleshooting may have

Keep testing until you are sure that you have found the right solution. If nothing works, you will need to rethink what the actual cause is.

Step 5: Implement the solution

Once you have accurately diagnosed the problem, found the solution, and tested it, then it is time to fix it. Even if your solution worked during testing, it is important to test it again. Ensure the asset is working the way it should. Note the steps taken for future reference.